

FD solver — the single teacher

wave/fd_solver.py: conservative leapfrog, harmonic-mean $E_{i+1/2}$, Mur ABCs

768 training labels

wave/ — **grid-to-grid FNO2d**

$[E, \rho, g, x, t] \rightarrow u(x, t)$ (whole field, one pass)

trained on 768 random materials + pulses

predicts u directly — IC learned from data

13.1M params · 3.2% val · 28–36% canonical @ 256^2

32-solve σ -bank

final/ML — **ansatz-conditioned FNO1d**

$[x, E, g, t] \rightarrow$ ansatz NN term (slice by slice)

one material; bank of 32 pulse widths σ_g

IC exact via the shared hard-constraint ansatz

2.53% best mean L_2 on exp1 · exp2/3 not yet run

bracket the design space: purely data-driven $\longleftrightarrow$ structure-sharing with the PINN formulation

accuracies answer different questions — do *not* rank 2.53% against 3.2%